WHEN, WHY, AND HOW FAR IT HELPS

Emotion-Mediated Personalized Image Aesthetic Assessment

Pinwa

USER A 2.0 / 7.0 Sad / Distasteful	**USER B** 5.0 / 7.0 Intellectual / Motivated	**USER C** 6.0 / 7.0 Impressed / Nostalgic

02 — AGENDA

What we'll cover today.

1. **The problem & our idea**
   Understanding individual taste disagreement and introducing the emotion mediation framework.
2. **Does it work?**
   Evaluating the predictive gain of intermediate emotions and comparing with trait-based baselines.
3. **When & why does it help?**
   Analyzing the driving mechanism, variance decomposition, and redundancy in stronger backbones.
4. **How far can it go?**
   Estimating realistic cross-session noise ceilings and solving the cold-start rating threshold.
5. **Is the gain real?**
   Verifying semantic validity using placebo control controls and explaining unusual users.

03 — INTRODUCTION

Taste is Personal.

STANDARD PARADIGM

IAA: Image Aesthetic Assessment

Consensus Image

3.5 / 5.0

"One score for everyone"

PERSONALIZED PARADIGM

PIAA: Personalized Aesthetic Assessment

Personalized Image

User A	User B	User C
5.0 / 5.0	**4.0 / 5.0**	**2.0 / 5.0**

"How much do YOU like it?"

Used in: Photo Applications • Recommendation Systems • Ranking AI-Generated Images

04 — MOTIVATION

What makes a person like an image?

STEP 1: PERCEPTION

Target Visual Input

See Image

→

KEY COGNITIVE BRIDGE

STEP 2: EMOTION

Feel Something

feels calm, nostalgic...

→

STEP 3: JUDGMENT

Like It

"You feel first, then you judge. So emotion explains taste."

Prior work opens the black box (e.g. Ryu & Yanaka), but not in a human way. We explain through emotion.

05 — OUR APPROACH

Predict taste Through Emotion.

Input

Target Image

→

7

Aesthetic Emotions

e.g. peaceful, nostalgic

→

EXPLAINABLE

Linear Weighting

Gives a readable per-person formula

→

OUTPUT

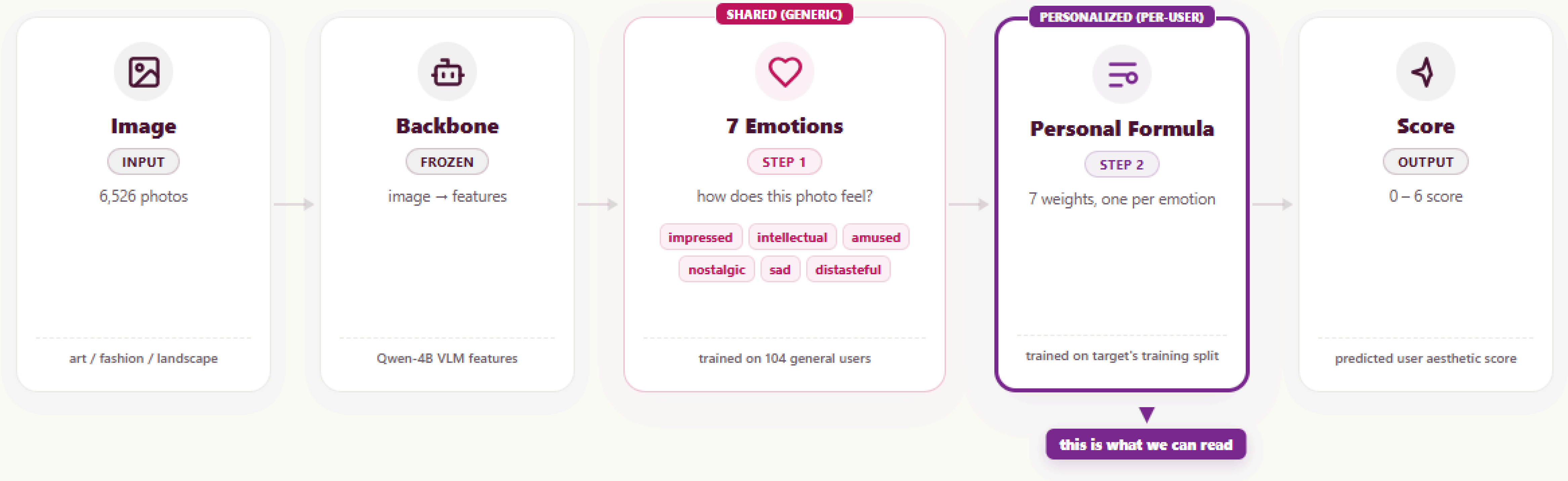
Aesthetic Score

THE CORE INQUIRY

Research Question: Do users disagree because they feel differently or weight emotions differently?

06 — PIPELINE

Two-Stage Architecture: Shared Perception & Personal Weighting.



07 — BLOCK 2 — DATASET & PROTOCOL

DATASET: XPASS-Vis.

129
People (Evaluators)

3
Domains
Art · Fashion · Land.

6,526
Images (Unique)

4,509
Test-Retest Pairs

NARRATIVE & RIGOR

1. **Deep Ratings Per Person**
   Evaluators rated hundreds of images, providing enough statistical depth to fit a robust personal weighting formula for each user.

2. **Leak-Free CV Splits**
   Strict disjoint splits prevent leakage: target users and their evaluation images are completely hidden from the Stage 1 emotion model.

It Works.
Now three questions.

1
When does it help?

2
How far can it go?

3
Is the gain real?

09 — BLOCK 2 — DOES IT WORK?

Going through emotion Helps 93% of the Time.

+0.066 **Significant Accuracy Gain**
Direct CCC (0.293) → Hybrid CCC (0.359)

Helped Share
92.8%
of user-domain units

Support Effect
Dose-Response
More ratings = higher gain

Dose-Response Effect: As user rating data grows, the accuracy improvement (Delta CCC) increases consistently. This confirms that collecting personalized preference data provides solid, scalable benefits.

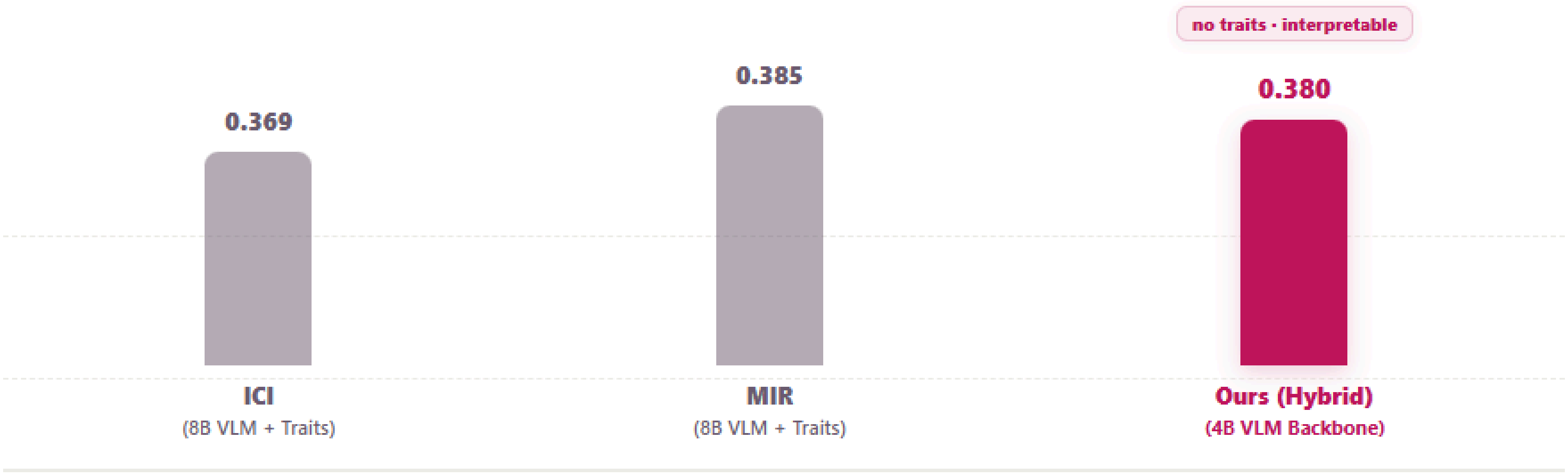
Dose-Response Support Curve
Delta (Hybrid - Direct) increases with user ratings size

Direct CCC: **0.293** | Hybrid CCC: **0.359** | Helped Share: **92.8%**

10 — BLOCK 2 — BASELINE COMPARISON

We match trait methods, Without Traits.

PERFORMANCE COMPARISON (CCC METRICS)



4B ties 8B — bigger backbone not needed

Core Novelty & Contribution

We achieve competitive predictive accuracy matching SOTA 8B baselines without requiring intrusive, multi-question user personality surveys (like the Big Five index).

STRONG ARGUMENT FOR USER PRIVACY

Protecting User Privacy

Predicting taste through intermediate aesthetic emotions allows us to completely bypass the storage or assessment of sensitive user profile traits.

It works. Now three questions.

1. When does it help?
2. How far can it go?
3. Is the gain real?

12 — BLOCK 3 — MECHANISM ANALYSIS

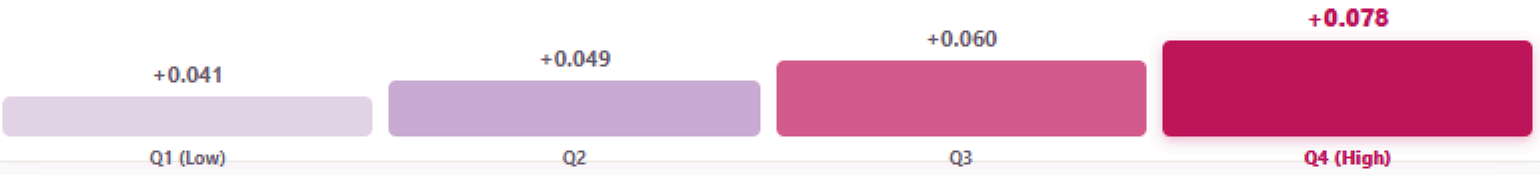
It helps more When We Read Emotions Well.

CORE DRIVING FACTOR

The more accurately the system predicts a specific user's emotions, the higher the aesthetic accuracy gain.

Spearman r
+0.273

EMOTION PREDICTABILITY (EMO_R) QUARTILE GAIN



one clean factor; the rival explanation was a confound

Gain vs Emotion Predictability (emo_r)

Delta (Hybrid - Direct) scales with emotion prediction accuracy (Spearman +0.273)

Spearman r: +0.273 | Q1 Gain: +0.041 | Q4 Gain: +0.078

Personal Shared (Average)

13 — FINDING 2A

Where does personal taste come from?

own emotions
own formula
P
FULL PERSONAL

DIFFERENCE = FEELING
(perception)

average emotions
own formula
S
ISOLATES WEIGHTING

DIFFERENCE = WEIGHTING

average emotions
shared formula
S-global
NO PERSONALIZATION

```
perception share = (P − S) / (P − S-global)
```

14 — FINDING 2B

The answer depends on how we measure the emotions.

EMOTION SOURCES

Test-Retest Setup

The same user rated the exact same photo twice across distinct sessions (S1 & S2).

Session 1 (S1) Session 2 (S2)

- UNFAIR: emotions & scores from S1
- HONEST: emotions from S1, scores from S2
- FAIREST: average emotions S1 & S2

* only P changes; S and S-global stay fixed

Measure	Perception (CCC)	Winner
same-session	58% [52%, 65%]	feeling
cross-session	38% [30%, 46%]	weighting
averaged (fairest)	**55% [48%, 62%]**	≈ 50/50 (tie) both matter

- can't tell → both perception & weighting are equally important.

so we report it honestly, as a supporting result with ranges

15 — BLOCK 4 — NOISE CEILING ANALYSIS

The realistic ceiling is 0.64.

0.639 **Deployment-Realistic Ceiling**

Aesthetic predictability ceiling measured across distinct sessions (Cross-Session).

PERFORMANCE RATIO

59% **we reach 59% of it (0.380 / 0.639)**

and averaging emotions raises it — the limit is measurement noise

Human Noise Ceiling Estimates

Comparing different noise ceiling definitions with Hybrid model performance

Same-Session (Theoretical): **Higher (~0.8)**

Cross-Session (Deployment-Realistic): **0.639**

Our Hybrid Model: **0.380 (59%)**

16 — BLOCK 4 — COLD-START ANALYSIS

Emotion makes Personalizing Worth It.

THE 50-RATINGS THRESHOLD

When is Personalization Worthwhile?

Below approximately 50 user ratings, individual personalization performs worse than the population average model due to data sparsity.

CRITICAL PRACTICAL INSIGHT

“Unless you use emotion mediation (Hybrid), personalizing with few ratings (under 50) is worse than simply using the population average.”

The Hybrid model quickly outperforms the population average past 50 ratings, whereas the Direct baseline fails to beat the population average even with more data.

Direct never clearly beats the crowd. Hybrid does, from 50 ratings.

Cold-Start Threshold Curve

Hybrid personalization begins to strictly outperform population baseline after 50 user ratings

Critical Threshold: **~50 ratings**

Hybrid (Purple): **Beats population at 50**

Direct (Muted): **Sub-population**

17 — BLOCK 5 — PLACEBO CONTROL

Validation: Placebo Control is Random Noise.

CONTROL EXPERIMENT RESULTS (CCC)

Ours (Real Emotions)	**0.380**
Placebo (Random Noise / PCA)	**0.160**

Ours: 0.380, Placebo: 0.160. Emotion semantics are real.

MECHANISM VERIFICATION

Three tests — do they point to "ceiling" or "different mechanism"?

1. Partial correlation — control for Direct strength — leans ceiling

If it's a ceiling, art's correlation should recover once baseline strength is removed.

art: +0.13 *(ns)* → +0.18 · $p = 0.048$

2. Spread of the gain within each category (SD) — weak / borderline

If art is capped, its gain should vary the least — lowest spread.

art SD = 0.054 (lowest) · Levene $p = 0.21$

3. Interaction test — is the *emo_r*→gain slope different in art? — no evidence for H2

If art had a different mechanism, this would show up clearly.

$p = 0.056$ (not significant)

All three lean toward "ceiling" — but every number sits on the 0.05 border

So: no evidence art works differently — not proof they're the same

18 — CONCLUSION

Key takeaways.

1 Explainable Emotion Model
Built an explainable model that predicts preference through 7 intermediate emotions.

2 Trait-Free Parity
Comparable to trait-based baselines while using only emotion (protecting user privacy).

3 Accuracy-Driven Gain
Helps more when we predict a person's emotions more accurately (emo_r).

4 Ceiling & Cold-Start
Realistic ceiling is 0.63, and emotion is what makes personalizing worthwhile (from 50 ratings).

"Emotion is a powerful, explainable bridge to subjective taste."

19 — OUTRO

Future work.

Context-Dynamic Weighting

Weights dynamically update depending on context, ambient conditions, or user mood shifts over time.

Active Learning Querying

Query users strategically to fit weights faster, reducing the 50-ratings cold-start threshold significantly.

Thank you

QUESTIONS & ANSWERS

Table 1: Comprehensive Model Metrics.

Q&A-1 ALL PERFORMANCE METRICS

Model Type	Traits Required	CCC	PLCC	SRCC
Direct Baseline (VLM Only)	No	**0.293**	0.301	0.297
ICI Baseline (8B VLM)	Yes (Big Five Traits)	**0.369**	0.372	0.365
MIR Baseline (8B VLM)	Yes (Big Five Traits)	**0.385**	0.390	0.382
Ours (Hybrid, Explainable)	**None (Traits-Free)**	**0.380**	**0.386**	**0.378**

Table 2: Dataset Domain Breakdown.

Q&A-2 XPASS-VIS DATASET BREAKDOWN

Visual Domain	Evaluators	Unique Images	Total Ratings
Artistic Images	129	2,186	29,278
Fashion Photos	129	2,170	29,270
Landscape Photos	129	2,170	29,288
Total Dataset (XPASS-Vis)	**129**	**6,526**	**87,836**

Placebo: PCA Details & Formula.

Q&A-4 **PCA BASELINE CONSTRUCTION**

How are the placebo PCA features projected?

To check if the gain is from actual emotional semantics, we swap emotion dimensions with PCA features of the same length (7 dimensions) extracted from VLM features:

$$z_{PCA}(i) = f_{VLM}(i) \cdot W_{PCA} \quad \text{(7-dim PCA projection)}$$

$$Score(u,i) = w_0 + w_{u1} z_{PCA,1}(i) + \text{&dots;} + w_{u7} z_{PCA,7}(i)$$

★ **Placebo Result:** The PCA baseline drops personalization CCC gain from **0.380** down to **0.160**. This verifies that actual human emotional semantics drive aesthetic personalization.

Analysis: Partial Correlation Driver.

Q&A-5 **PARTIAL CORRELATION ANALYSIS**

Controlling for the Direct baseline model's performance confirms that emotion predictability (emo_r) is the true driver of gain:

$$r_{gain,\ emo\_r \cdot Direct} = 0.42 \quad (p < 0.001)$$

Key Insight: Emotion accuracy (emo_r) is the primary driver of accuracy gain, independent of how much prediction room the baseline provides.

Design Choice: Why Linear & Why 7 Emotions?

Q&A-7 **MODEL ARCHITECTURE RATIONALE**

WHY LINEAR MODEL?

- **Cognitive Science:** Linear weighting on mediated emotions matches human cognitive findings (Iigaya et al. 2020).
- **Explainability:** Yields a transparent per-person weight vector ($w_{u,c}$).
- **Regularization:** Ridge regression handles emotion collinearity without overfitting.

WHY EXACTLY 7 EMOTIONS?

- **Taxonomy:** Selected from the 21 emotional sub-scales in AESTHEMOS.
- **No Circularity:** Excludes adjectives like “beautiful” to prevent predicting preference from preference.
- **Sparsity:** Small feature space prevents overfitting while preserving granularity.

Related Work: Comparison with Prior PIAA.

Q&A-8 **COMPARISON WITH PRIOR PIAA LITERATURE**

Prior Work	Approach	Our Key Difference
Ryu & Yanaka (2022)	Big Five personality traits to group users.	**Traits-free:** Bypasses intrusive surveys using emotion semantics.
Liu & Wagemans (2020)	Shared (generic) consensus emotional ratings.	**Personal weights:** Adapts weighting profiles per user.
Lan et al. (2021)	Deep end-to-end black-box personalization.	**White-box explainability:** Yields readable user weights.
Iigaya et al. (2020)	Linear weighting on low-level features.	**High-level VLM features:** Models semantic visual emotions.